

Gallbladder and Gallstones

Gallstones and cholecystitis, cholesterol in bile (ABCG8, ABCB4), bilirubin and Gilbert syndrome (UGT1A1*28), and gallbladder cancer

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F's genes (genotyping chip + local imputation). It is not a diagnosis or a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The largest common gallstone gene is in Sérgio F's favour: **ABCG8 D19H is absent** (G/G), as is **HNF4A T130I** (C/C), and LRBA and MARCHF8 carry no risk alleles. The two most recent genomic cholelithiasis scores are at percentiles **32.4** and **33.8**, the 13-variant score at percentile 6.3, and the burden of 124 gallstone loci at percentile 32.5. The consensus of the scores with good coverage is at percentile **51.6** (typical) and that of the genomic scores at percentile 33.8.

Unfavourable. An older genomic score for cholelithiasis and cholecystitis is at percentile **88.8**, and two small scores at percentiles 69.4 and 71.6. There are several small-effect risk alleles in cholesterol and bile salt transport: **ABCG8 rs4245791** (T/T, 2 alleles), **ABCB4** (G/G, 2 alleles), **CYP7A1** (C/C), **TTC39B** (A/A) and a heterozygous **ABCB11 V444A**. On top of this come a heterozygous **UGT1A1*28** (Gilbert, C/T), which raises bilirubin, and the **PiZ allele of alpha-1 antitrypsin** (C/T), which is also associated with more gallstones. The cholecystitis loci burden is at percentile 91.9, but with only 11 loci.

What it means. The genetic risk of gallstones is typical: the consensus of the scores sits in the middle of the distribution and ABCG8 D19H, the largest common factor, is absent. The risk alleles present have small effects (about 1.1–1.3× each). What matters most for gallstones is behaviour: rapid fat loss, long fasts with no fat at all, little fibre and little physical activity. Genetically slightly higher bilirubin (Gilbert) has an upside: it is a natural antioxidant and is associated with less cardiovascular disease. Gallbladder cancer is rare; the two ABCB4 markers read here were discovered in very high-risk groups and count for little in this context.

2. How to read this report

- **Data:** genotypes from Sérgio F's chip (GRCh38, + strand) and, when the SNP is not on the chip, locally imputed genotypes (statistical imputation, public panel of 30x whole genomes) with quality DR2 $\geq$ 0.8. Below this threshold the SNP is treated as not found in the data.
- **Key variants:** 19 SNPs selected from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 16 from the chip, 3 imputed and 0 not found.
- **Strand:** all genotypes are on the + strand of GRCh38. The published effect allele is compared with the observed alleles; 3 SNPs needed conversion to the complementary strand and 1 is palindromic (A/T or C/G), resolved by the reference allele and the position. Where the literature uses the gene strand, the letters may differ from those in this report.
- **Polygenic scores (PGS):** 11 PGS Catalog scores calculated locally with 3,577,484 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentile ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **GWAS loci burden:** for each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in the GWAS Catalog (1 row per SNP, the most significant) are added up as a simple count of effect alleles (each allele counts 1). Sérgio F's sum is compared with that of the reference population ($n = 633$) using exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium with each other, so the burden is a complement to, not a substitute for, the polygenic scores. 1063 loci, 832 with genotype.
- **Judgement:** each result is classed as Favourable or Unfavourable (or typical / no judgement). A genotype is a probability, not a destiny; almost all effects are small and add to lifestyle.
- **Evidence:** Strong = replicated and with a clear clinical effect; Moderate = consistent GWAS/meta-analyses, modest effect; Weak = small or purely mechanistic studies.

3. Polygenic scores

Consensus by trait. For each group, the consensus is the **median of the percentiles of the scores with effect-size coverage $\geq 75\%$** , all calibrated in the same reference population ($n = 633$). The genomic column is the median of the scores with $\geq 100,000$ variants. Scores with coverage $< 75\%$ are shown in the tables but do not enter the consensus.

Group	Consensus (percentile)	Scores with coverage $\geq 75\%$	Genomic scores (percentile)	Genomic scores
Gallstones	51.6	6 of 7	33.8	3
Cholecystitis	—	0 of 1	—	0
Bilirubin	71.4	3 of 3	72.2	2

For gallstones and cholecystitis, a higher percentile means more risk. For bilirubin, the percentile is descriptive only (no judgement): more bilirubin slightly increases pigment gallstones, but it is an antioxidant. Two 2021 scores (183,000 variants) have 41% coverage and are left out of the consensus.

Gallstones

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Cholelithiasis (genomic, 2025)	PGS005150 (Jung 2025, 908,000 variants)	886,908 of 908,465 (coverage 97%)	32.4	TYPICAL	typical
Cholelithiasis (genomic)	PGS004476 (Jung 2024, 1.06 million variants)	1,025,639 of 1,059,939 (coverage 96%)	33.8	TYPICAL	typical
Cholelithiasis and cholecystitis (genomic)	PGS002072 (Privé 2022, 429,000 variants)	421,201 of 428,587 (coverage 98%)	88.8	MORE LIKELY	unfavourable
Gallstones (genomic)	PGS000705 (Sinnott-Armstrong 2021, 183,000 variants)	74,380 of 183,458 (coverage 41%)	29.9	TYPICAL	typical
Cholelithiasis and cholecystitis	PGS001861 (Privé 2022, 2,059 variants)	2,012 of 2,059 (coverage 98%)	69.4	TYPICAL	typical
Gallstones	PGS001256 (Tanigawa 2022, 876 variants)	772 of 876 (coverage 91%)	71.6	TYPICAL	typical
Cholelithiasis	PGS012582 (PGS Catalog 2026, 13 variants)	12 of 13 (coverage 96%)	6.3	LESS LIKELY	favourable

Cholecystitis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Cholecystitis (genomic)	PGS000707 (Sinnott-Armstrong 2021, 183,000 variants)	74,324 of 183,329 (coverage 41%)	23.4	TYPICAL	typical

Bilirubin

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Total bilirubin (genomic)	PGS003524 (Ding 2023, 980,000 variants)	972,627 of 979,739 (coverage 100%)	74.9	TYPICAL	no judgement
Total bilirubin (genomic)	PGS002160 (Privé 2022, 120,000 variants)	118,548 of 120,068 (coverage 100%)	69.5	TYPICAL	no judgement
Total bilirubin	PGS000697 (PGS Catalog 2021, 1,159 variants)	1,061 of 1,159 (coverage 96%)	71.4	TYPICAL	no judgement

4. Key variants by topic

Replicated variants of the bile transporters (ABCG5/8, ABCB4, ABCB11), bile acid synthesis (CYP7A1), bilirubin (UGT1A1) and liver genes, plus two gallbladder cancer markers. Published alleles with a protective effect were inverted to the risk direction.

Cholesterol in bile (ABCG5/8, ABCB4, CYP7A1)

Cholesterol precipitates when bile has too much cholesterol or too few bile salts and phospholipids.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ABCG8	rs11887534	D19H	G/G	chip	C (0 copies)	favourable	D19H: the C allele (His19) increases the output of cholesterol from the liver to the bile; ~2× more gallstones per copy (the largest common locus)	Strong
ABCG8	rs4245791		T/T	chip	T (2 copies)	unfavourable	Biliary cholesterol ABCG5/8 transporter (signal independent of D19H) [GWAS Catalog: Gallstone disease, allele T, OR 1.28, p=5E-31, PMID 27094239]	Moderate
ABCB4	rs4148805		G/G	chip	G (2 copies)	unfavourable	Phosphatidylcholine transporter into bile (MDR3): less phospholipid makes bile more lithogenic [GWAS Catalog: Cholelithiasis, allele A, beta 0.139 [0.12-0.16] unit decrease; risk/increase allele = G, the opposite, p=2E-36, PMID 34594039]	Moderate
ABCB11	rs2287622	V444A	A/G	chip	C (1 copy)	unfavourable (mild)	V444A of the bile salt export pump (BSEP): the C allele (Ala444) reduces BSEP expression; more cholestasis and more gallstones	Moderate
TM4SF4	rs4681515		A/G	imputed DR2=0.97	A (1 copy)	unfavourable (mild)	TM4SF4/TM4SF1 locus (gallbladder and liver epithelium) [GWAS Catalog: Cholelithiasis, allele G, beta 0.125 [0.11-0.14] unit decrease; risk/increase allele = A, the opposite, p=9E-53, PMID 34594039]	Moderate
CYP7A1	rs983812		C/C	imputed DR2=1.00	C (2 copies)	unfavourable	CYP7A1 region, the enzyme that converts cholesterol into bile acids [GWAS Catalog: Gallstone disease, allele T, OR 0.90; risk/increase allele = C, the opposite, p=1E-42, PMID 34651315]	Moderate

Bilirubin and Gilbert syndrome (UGT1A1)

Conjugation of bilirubin in the liver; Gilbert syndrome is benign and common.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
UGT1A1	rs887829	*28 (marker)	C/T	chip	T (1 copy)	unfavourable (mild)	UGT1A1*28 marker (Gilbert syndrome): the T allele lowers the conjugation of bilirubin; more bilirubin and more pigment gallstones (~1.2× per copy), but bilirubin is antioxidant	Strong
UGT1A1	rs6742078		G/T	chip	T (1 copy)	unfavourable (mild)	Total bilirubin (UGT1A1) main locus: the T allele raises bilirubin and is associated with more gallstones	Strong

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SLCO1B1	rs4149056	V174A	T/T	chip	C (0 copies)	no judgement	SLCO1B1*5 (V174A): less hepatic uptake of bilirubin and statins	Moderate

Liver, metabolism and other gallstone loci

Liver transcription factors, lipid metabolism and the microbiome.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HNF4A	rs1800961	T130I	C/C	chip	T (0 copies)	favourable	T130I of HNF4A (liver transcription factor): ~1.3x more gallstones	Moderate
SERPINA1	rs28929474	PiZ (E342K)	C/T	chip	T (1 copy)	unfavourable (mild)	PiZ allele of alpha-1 antitrypsin (E342K): ~1.4x more gallstones per copy (link to liver and lung)	Strong
LRBA	rs2290846		G/G	imputed DR2=0.99	A (0 copies)	favourable	LRBA/MAB21L2 locus [GWAS Catalog: Gallstone disease, allele A, OR 1.13, p=5E-46, PMID 34651315]	Moderate
GCKR	rs1260326	P446L	T/C	chip	C (1 copy)	unfavourable (mild)	P446L of glucokinase regulatory protein: triglycerides and hepatic fat [GWAS Catalog: Gallstone disease, allele C, OR 1.08, p=1E-23, PMID 34651315]	Moderate
TTC39B	rs686030		A/A	chip	A (2 copies)	unfavourable	TTC39B: HDL and biliary cholesterol metabolism [GWAS Catalog: Gallstone disease, allele A, OR 1.14, p=4E-30, PMID 34651315]	Moderate
FUT2	rs601338	W143X	G/A	chip	A (1 copy)	unfavourable (mild)	FUT2 W143X (non-secretor status): microbiome and bile acids [GWAS Catalog: Gallstone disease, allele A, OR 1.09, p=5E-32, PMID 34651315]	Moderate
HNF1A	rs2393791		C/T	chip	T (1 copy)	unfavourable (mild)	HNF1A (liver transcription factor) [GWAS Catalog: Gallstone disease, allele T, OR 1.09, p=8E-27, PMID 34651315]	Weak
MARCHF8	rs970548		A/A	chip	C (0 copies)	favourable	MARCHF8 locus [GWAS Catalog: Gallstone disease, allele C, OR 1.11, p=2E-35, PMID 34651315]	Weak

Gallbladder and biliary tract cancer

Markers discovered in very high-risk groups; informative reading.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ABCB4	rs1558375		T/T	chip	A (2 copies)	unfavourable	ABCB4 and gallbladder cancer: ~1.5x per copy in high-risk population studies	Moderate
ABCB1/ABCB4	rs17209837		T/T	chip	A (2 copies)	unfavourable	ABCB1/ABCB4 region and gallbladder cancer: ~1.6x per copy	Moderate

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 5 \times 10^{-8}$) for gallstones, cholecystitis, bilirubin and gallbladder cancer. Alleles are oriented towards more risk (or more bilirubin) and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F's effect alleles	Calibrated variants	Percentile	Judgement
Gallstones	124 (104)	91 of 208	104	32.5	typical
Cholecystitis	15 (11)	12 of 22	11	91.9	unfavourable
Total bilirubin	921 (714)	715 of 1428	713	71.2	no judgement
Gallbladder and biliary tract cancer	3 (3)	4 of 6	3	66.7	typical

6. Favourable

- **ABCG8 D19H absent** (G/G): without the largest common genetic factor for cholesterol gallstones (about 2× per copy). **Strong.**
- **Recent genomic cholelithiasis scores** at percentiles **32.4** and **33.8**; 13-variant score at percentile 6.3; loci burden at percentile 32.5. **Moderate.**
- **HNF4A T130I absent** (C/C); LRBA (G/G) and MARCHF8 (A/A) without risk alleles. **Moderate.**
- **Genetically slightly higher bilirubin** (heterozygous Gilbert; scores at percentiles 74.9, 69.5 and 71.4): an endogenous antioxidant, associated with less atherosclerosis. **Moderate.**

7. Unfavourable

- **Genomic score for cholelithiasis and cholecystitis (2022)** at percentile **88.8**; small scores at percentiles 69.4 and 71.6. The most recent scores do not confirm it. **Weak to moderate.**
- **ABCG8 rs4245791** T/T and **ABCB4** G/G (2 alleles each), **CYP7A1** C/C and **TTC39B** A/A (2 alleles), **ABCB11 V444A** A/G, and GCKR, FUT2, HNF1A and TM4SF4 with 1 allele. Effects of about 1.1–1.3× each. **Moderate.**
- **Heterozygous UGT1A1*28** (C/T; rs6742078 G/T): more pigment gallstones (about 1.2×) and caution with irinotecan and atazanavir. Strong (genotype).
- **Heterozygous SERPINA1 PiZ** (C/T): about 1.4× more gallstones and a cofactor for liver disease. Strong (genotype).
- **Cholecystitis**: loci burden at **91.9** (11 loci). **Weak.**
- **ABCB4 gallbladder cancer markers** in homozygosity (T/T, T/T), but the burden of these loci is typical (66.7) and the cancer is rare. **Weak.**

8. Mechanisms of ageing

Mapping onto the 84 ageing mechanisms of Sérgio F's immortality goals (only the mechanisms touched by this report).

Mechanism	Genetic finding	Meaning
Metabolic Homeostasis Decline	Cholesterol and bile salt transport (ABCG8 D19H absent; ABCB4, ABCB11 and CYP7A1 with risk alleles); gallstone consensus at percentile 51.6	Typical
Oxidative Stress	Genetically slightly higher bilirubin (heterozygous UGT1A1*28): an endogenous antioxidant	Favourable
Protein Aggregation	Heterozygous alpha-1 antitrypsin PiZ: the Z protein aggregates in the liver (a cofactor for fibrosis and gallstones)	Unfavourable (mild)
Chronic Inflammation	Cholecystitis with loci burden at percentile 91.9 (few loci); gallbladder inflammation almost always starts from gallstones	Conditional
Dysbiosis	Heterozygous FUT2 (secretor) and bile acids: bile shapes the microbiome and the microbiome transforms bile acids	Typical
Insulin Resistance	Insulin resistance increases cholesterol in bile; GCKR and HNF1A with 1 allele	Conditional
Somatic Mutation Accumulation	Gallbladder cancer: ABCB4 markers in homozygosity, typical burden; rare cancer	Typical

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **During rapid fat loss** (for example with GLP-1/GIP agonist medicines), keep regular meals with some fat so the gallbladder contracts, keep fibre high and avoid long fasts with zero fat. If the loss is fast, discuss preventive ursodeoxycholic acid with the doctor. Suggestion (not applied).
2. **Abdominal ultrasound** only if there is pain under the right ribs after meals, or by taking advantage of an ultrasound already requested for another reason. Gallstones without symptoms do not need treatment. Suggestion (not applied).
3. **Bilirubin in blood tests**: a slightly high total with a normal direct fraction is expected in Gilbert syndrome and rises with fasting and acute illness; it does not require investigation. Suggestion (not applied).
4. **Drug card**: keep UGT1A1*28 on record (irinotecan and atazanavir). Suggestion (not applied).
5. **Already in the plan and protective**: high fibre, coffee, daily exercise, zero alcohol and glucose control. Statins also slightly lower the risk of gallstones. Suggestion (not applied).

10. Limitations

- The cholelithiasis scores disagree with each other (from 6.3 to 88.8); the consensus and the most recent genomic scores sit in the middle, so the result is read as typical.
- The gallbladder cancer markers were discovered in very high-risk groups and the allele may be on the opposite strand; the reading is informative.
- Gilbert syndrome is inferred from the rs887829 marker (the TA repeat in the promoter is not read directly).
- Gallstones depend heavily on non-genetic factors (diet, rapid fat loss, hormones, physical activity).
- The loci burden adds up SNPs in linkage disequilibrium; it is descriptive.
- Conclusions from genes only; this is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F's chip (GRCh38, + strand). Missing SNPs come from local statistical imputation (public panel of 30x whole genomes), used only with $DR2 \geq 0.8$. No genetic data left the analysis computer.
- **Alleles and strand:** the published effect allele is compared with the alleles of the imputed VCF; for palindromic SNPs the + strand is assumed (the GWAS Catalog and PGS Catalog convention). Published alleles with a protective effect were inverted to the risk direction.
- **Polygenic scores:** GRCh38 harmonised files from the PGS Catalog (11 scores: cholelithiasis, cholecystitis and bilirubin). Score = $\sum \text{coefficient} \times \text{dosage}$. The percentile is empirical in the reference population ($n = 633$), using the same variants. Consensus = median of the scores with effect-size coverage $\geq 75\%$.
- **Loci burden:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), 4 sets, alleles oriented by the sign of the published effect, with empirical calibration in the same reference population.
- **Interpretation:** Buch 2007 (ABCG8 D19H), Ferkingstad 2018 and Fairfield 2022 (gallstone GWAS), Bosma 1995 (UGT1A1*28), Ferkingstad 2018 (SERPINA1 Z and gallstones), Mhatre 2017 (gallbladder cancer), PGS Catalog, GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Strand	Interpretation
Cholesterol in bile (ABCG5/8, ABCB4, CYP7A1)	ABCG8	rs11887534	G/G	chip	C	0	+ strand (palindromic; + assumed)	without the risk allele
Cholesterol in bile (ABCG5/8, ABCB4, CYP7A1)	ABCG8	rs4245791	T/T	chip	T	2	+ strand	2 copies of the risk allele
Cholesterol in bile (ABCG5/8, ABCB4, CYP7A1)	ABCB4	rs4148805	G/G	chip	G	2	+ strand	2 copies of the risk allele
Cholesterol in bile (ABCG5/8, ABCB4, CYP7A1)	ABCB11	rs2287622	A/G	chip	C	1	complement (– strand)	1 copy of the risk allele
Cholesterol in bile (ABCG5/8, ABCB4, CYP7A1)	TM4SF4	rs4681515	A/G	imputed DR2=0.97	A	1	+ strand	1 copy of the risk allele
Cholesterol in bile (ABCG5/8, ABCB4, CYP7A1)	CYP7A1	rs983812	C/C	imputed DR2=1.00	C	2	+ strand	2 copies of the risk allele
Bilirubin and Gilbert syndrome (UGT1A1)	UGT1A1	rs887829	C/T	chip	T	1	+ strand	1 copy of the risk allele
Bilirubin and Gilbert syndrome (UGT1A1)	UGT1A1	rs6742078	G/T	chip	T	1	+ strand	1 copy of the risk allele
Bilirubin and Gilbert syndrome (UGT1A1)	SLCO1B1	rs4149056	T/T	chip	C	0	+ strand	0 copies of the effect allele
Liver, metabolism and other gallstone loci	HNF4A	rs1800961	C/C	chip	T	0	+ strand	without the risk allele
Liver, metabolism and other gallstone loci	SERPINA1	rs28929474	C/T	chip	T	1	+ strand	1 copy of the risk allele
Liver, metabolism and other gallstone loci	LRBA	rs2290846	G/G	imputed DR2=0.99	A	0	+ strand	without the risk allele
Liver, metabolism and other gallstone loci	GCKR	rs1260326	T/C	chip	C	1	+ strand	1 copy of the risk allele
Liver, metabolism and other gallstone loci	TTC39B	rs686030	A/A	chip	A	2	+ strand	2 copies of the risk allele
Liver, metabolism and other gallstone loci	FUT2	rs601338	G/A	chip	A	1	+ strand	1 copy of the risk allele
Liver, metabolism and other gallstone loci	HNF1A	rs2393791	C/T	chip	T	1	+ strand	1 copy of the risk allele
Liver, metabolism and other gallstone loci	MARCHF8	rs970548	A/A	chip	C	0	+ strand	without the risk allele
Gallbladder and biliary tract cancer	ABCB4	rs1558375	T/T	chip	A	2	complement (– strand)	2 copies of the risk allele
Gallbladder and biliary tract cancer	ABCB1/ ABCB4	rs17209837	T/T	chip	A	2	complement (– strand)	2 copies of the risk allele